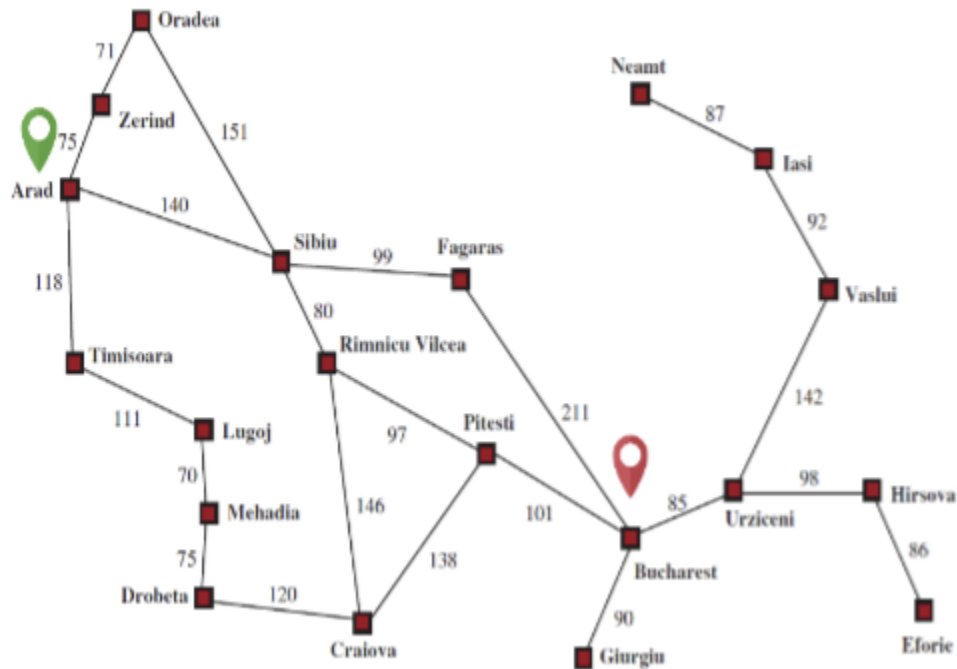


CS 4720 HW 3



h_{SLD} : straight-line distance

Arad	366	Mehadia	241
Bucharest	0	Neamt	234
Craiova	160	Oradea	380
Drobeta	242	Pitesti	100
Eforie	161	Rimnicu Vilcea	193
Fagaras	176	Sibiu	253
Giurgiu	77	Timisoara	329
Hirsova	151	Urziceni	80
Iasi	226	Vaslui	199
Lugoj	244	Zerind	374

1) Starting point: Lugoj (which means that g is 0) and we know that h is 244

Let's first consider the 2 neighbors of Lugoj (Mehadia and Timisoara) and the distance between Lugoj and Mehadia and Timisoara.

Timisoara:

Cost/g: 111

h: 329

$f = g + h = 329 + 111 = 440$

Mehadia:

Cost/g: 70

h: 241

$f = g + h = 70 + 241 = 311$

Now, we have Mehadia and Timisoara in our visited list and we already know that $f(\text{Lugoj})$ is 244

Visited: {Mehadia(311), Timisoara(440)}

Since $f(\text{Mehadia}) < f(\text{Timisoara})$, A* search will expand from $f(\text{Mehadia})$, so let us now take a look at its neighbor Drobeta.

Drobeta:

Cost/g: $70+75=145$

h: 242

$f = g + h = 242 + 145 = 387$

Now we have Drobeta and Timisoara in our visited list, and we already know that $f(\text{Lugoj})$ and $f(\text{Mehadia})$ are 244 and 311, respectively.

Visited: {Drobeta(387), Timisoara(440)}

Since $f(\text{Drobeta}) < f(\text{Timisoara})$, A* search will expand from $f(\text{Drobeta})$, so let us now take a look at its neighbor Craiova.

Craiova:

Cost/g: $145+120 = 265$

h: 160

$f = g + h = 160 + 265 = 425$

Now we have Craiova and Timisoara in our visited list, and we already know that $f(\text{Lugoj})$, $f(\text{Mehadia})$ and $f(\text{Drobeta})$ are 244, 311 and 387 respectively.

Visited: {Craiova(425), Timisoara(440)}

Since $f(\text{Craiova}) < f(\text{Timisoara})$, A* search will expand from $f(\text{Craiova})$, so let us now take a look at its neighbors Rimnicu Vilcea and Pitesti.

Rimnicu Vilcea

Cost/g: $265+146 = 411$

h: 193

$f = g + h = 411 + 193 = 604$

Pitesti

Cost/g: $265+138 = 403$

h: 100

$f = g + h = 403 + 100 = 503$

Now we have Rimnicu Vilicea, Pitesti and Timisoara in our visited list, and we already know that $f(\text{Lugoj})$, $f(\text{Mehadia})$, $f(\text{Drobeta})$ and $f(\text{Craiova})$ are 244,311,387 and 425 respectively.

Visited:{Rimnicu Vilicea(604),Pitesti(503), Timisoara(440)}

Since $f(\text{Timisoara}) < f(\text{Pitesti}) < f(\text{Rimnicu Vilicea})$, A* search will expand from $f(\text{Timisoara})$, so let us now take a look at its neighbor Arad, since Lujog has already been visited.

Arad

Cost/g: $111+118 = 229$

h: 366

$f = g + h = 366 + 229 = 595$

Now we have Rimnicu Vilicea, Pitesti and Arad in our visited list, and we already know that $f(\text{Lugoj})$, $f(\text{Mehadia})$, $f(\text{Drobeta})$, $f(\text{Craiova})$ and $f(\text{Timisoara})$ are 244,311,387,425 and 440 respectively.

Visited:{Rimnicu Vilicea(604),Pitesti(503), Arad(595)}

Since $f(\text{Pitesti}) < f(\text{Arad}) < f(\text{Rimnicu Vilicea})$, A* search will expand from $f(\text{Pitesti})$, so let us now take a look at its neighbor Bucharest and Rimnicu Vilicea.

Bucharest:

Cost/g: $403+101 = 504$

h: 0

$f = g + h = 504 + 0 = 504$

Rimnicu Vilicea

Cost/g: $403 + 97 = 500$

h: 193

$f = g + h = 500 + 193 = 693$ (which is more than its previous value 604, so it does not get updated).

Now we have Bucharest, Arad, and Rimnicu Vilicea in our visited list, and we already know that $f(\text{Lugoj})$, $f(\text{Mehadia})$, $f(\text{Drobeta})$, $f(\text{Craiova})$, $f(\text{Timisoara})$, and $f(\text{Pitesti})$ are 244, 311, 387, 425, 440, and 503 respectively.

Visited: {Rimnicu Vilicea(604), Arad(595), Bucharest(504)}

Since we have reached Bucharest and reached our goal, we can define the final path as follows:

Lugoj $\rightarrow$ Mehadia $\rightarrow$ Drobeta $\rightarrow$ Craiova $\rightarrow$ Timisoara $\rightarrow$ Pitesti $\rightarrow$ Bucharest

2a) In order to find the value for which this function is complete, we need to make sure that the value of the g coefficient is positive, which means that:

$$\begin{aligned} 2-w &> 0 \\ 2 &> w \end{aligned}$$

So, that's the upper bound for w. Plus, since we usually keep $w \geq 0$, we can say that 0 is the lower bound on w, hence we can say that the value of w lies somewhere in this interval, $[0, 2)$, or basically it will be 0 or 1.

b) In order to determine for what values of w, this function will be optimal, we need to make sure that the coefficients of both g and h are 1. In order to do that, let us divide the function by w.

$$f(n) = ((2-w)/w) g(n) + h(n)$$

Now, since we know that the coefficient of g should also be 1, we can do the following calculations:

$$\begin{aligned} (2-w)/w &= 1 \\ 2-w &= w \\ 2 &= 2w \\ 1 &= w \end{aligned}$$

Hence, we can say that for $w=1$, this function will be optimal.

c) Let us plug all these values of w into the function to find the answers to this question:

$$\begin{aligned} w &= 0: \\ f(n) &= 2g(n) \end{aligned}$$

Since it shows behavior similar to that of Uniform Cost Search, where $f(n) = g(n)$, we can say that we are performing Uniform Cost Search over here.

w=1:

$$f(n) = g(n) + h(n)$$

Since it shows behavior similar to that of A* search, where $f(n) = g(n) + h(n)$, we can say that we are performing A* search over here.

w=2:

$$f(n) = 2h(n)$$

Since it shows behavior similar to that of Greedy Best First Search, where $f(n) = h(n)$, we can say that we are performing Greedy Best First Search over here.

3a) Let us assume that vehicles are being placed one by one, so that means that the vehicle 1 has n^2 choices, vehicle 2 has $n^2 - 1$ choices, vehicle 3 $n^2 - 2$ has choices, which means that vehicle n will have $n^2 - (n - 1)$ choices.

Now, let us multiply all these terms and see what happens:

$$n^2 \times n^2 - 1 \times n^2 - 2 \times n^2 - 3 \times \dots (n^2 - (n-1)) = n^2! / (n^2 - n)!$$

We know that $n^2! = n^2 - 1 \times n^2 - 2 \times n^2 - 3 \times \dots$, which leaves us with extra terms such as terms such as $(n^2 - n - 1) \times (n^2 - n - 2) \times \dots$, which makes it equal to $(n^2 - n)!$, so that's why divide $n^2!$ by $(n^2 - n)!$, so that we can remove the extra terms.

b) Since the n distinct vehicles have 5 choices (up, down, stay, left, and right), which means that we 5^n have actions in total, hence we can say that the branching factor is $O(5^n)$.

c) Since there are no vehicles on the grid, we can say that the number of moves to reach from one point to the other one will exactly be equal to the Manhattan distance between the points, hence we can calculate the number of moves in the following way:

$$h_i = |x_i - (n-i+1)| + |y_i - n|$$

di) Since we are dealing with the number of moves required to reach the goal state, summing up all the moves for each car will overestimate the true cost, and hence it is not admissible.

ii) By looking at question 2, where we get the range of w as $0 \leq w < 2$, and then we obtain the optimal solution for w=1, which is still less than the maximum value 2, we can say that it is not admissible, as it is overestimating the true cost

iii) Similar to part (ii), we know that the range of w to complete the function is 0 and 1, where 0 is the minimum value, which means that it is admissible, since it's not overestimating the true cost.